

Information Technologies for Industrial Engineers

Fullstack Application

Setup Database

- Delete existing database container and volume.
- Run `docker run -d --name my-postgres -e POSTGRES_PASSWORD=1234 -v pgdata:/var/lib/postgresql -p 5432:5432 postgres:latest`

Note that this command specify the password of `1234` for the database. You can change it to your own password.

Prepare Database

In Dbeaver

```
CREATE DATABASE app_db;
```

```
CREATE TABLE IF NOT EXISTS  
  todos (  
    id serial PRIMARY KEY,  
    title VARCHAR(50) NOT NULL,  
    completed BOOLEAN NOT NULL DEFAULT FALSE,  
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()  
  );
```

Note the database name is `app_db` and the table name is `todos`.

Backend Starter

- `git clone https://github.com/it-for-ie-69/backend-todo`
- `cd backend-todo`
- `git init` (Optional)
- Create `.env` file from `.env.example` and set the database password to your own password.
- `pnpm i`
- `pnpm run dev`

Frontend Starter

- `git clone https://github.com/it-for-ie-69/frontend-todo.git`
- `cd frontend-todo`
- `git init` (Optional)
- `pnpm i`
- `pnpm run dev`

Note for Frontend

These are steps that I did in addition to the usual `pnpm create vite@latest` steps.

- `pnpm i axios` *(for making HTTP requests to the backend)*
- Create `.env` file and add `VITE_API_URL=http://localhost:3000` to it.

CORS Issues

- CORS stands for Cross-Origin Resource Sharing.
- CORS is a security feature implemented **by browsers** to restrict web pages from making requests to a different domain than the one that served the web page.
 - `http://localhost:5173` (frontend) is different from `http://localhost:3000` (backend).
- By default, **browsers** block cross-origin requests for security reasons. This is why you are getting CORS errors when trying to make requests from your frontend to your backend.

Test

- Backend that allows CORS requests
 - `fetch("https://jsonplaceholder.typicode.com/posts/1")`
- Backend that does not allow CORS requests
 - `fetch("https://google.com")`

What?

- You can type `https://google.com` in the browser address bar and it works.
- However, if you try to fetch `https://google.com` from your frontend code, it will fail due to CORS policy.
- This is because the typing in the address bar is a **direct request** from the browser, while the fetch request is a **cross-origin request** from your frontend code.

Fixing CORS in Backend

- To fix CORS issues in your backend, you need to enable CORS on your backend server.
 - `pnpm i cors`
 - `pnpm i -D @types/cors`

```
import cors from "cors";

//Intializing the express app
const app = express();
app.use(express.json());

// Enable CORS for all routes
app.use(cors());
```

Congratulations!

| You are now a fullstack developer!